A+ Core 1

Exam 220-1201

A+ 220-1201

Module 1

What Does an IT Specialist Do?

Learning Outcomes

Prepare for A+ Certification by:

- Describing what an IT specialist is and their responsibilities
- Describing the skills an IT specialist needs
- Describing the role of certifications for an IT specialist

Lesson 1.1

The Hero of Problem Solving

Role of an IT Specialist

Image © 123RF.com.

- Problem Solver
  - Responsibilities
- Range of Issues
  - Ever-changing daily tasking- No routine

Skills and Abilities (Slide 1 of 2)

- Problem Solving
  - Use a consistent process
- Communication
  - Written
  - Oral
  - Effective
- Organization
  - Clean workspace = easier to locate tools and parts
  - Organize thoughts that then are communicated to stakeholders

Image © 123RF.com.

Skills and Abilities (Slide 2 of 2)

Image © 123RF.com.

- Technical Knowledge
  - Hardware
  - Software- including operating systems and applications
  - Networking
  - Cloud
  - Security

Lesson Review: The Hero of Problem Solving

Role of an IT Specialist

Skills and Abilities

Lesson 1.2

The Troubleshooting Methodology

Best Practice Methodology

Determine cause, symptoms and consequences

Priority and severity

- Critical systems and most severe issues need to be addressed first

CompTIA Troubleshooting Methodology

- Identify the problem
- Establish a theory of probable cause
- Test the theory
- Establish a plan of action and implement the solution
- Verify full system functionality
- Document findings, actions, outcomes

Identify the Problem

Gather information from the user

- What are the exact error messages appearing on the screen or coming from the speaker?
- Is anyone else experiencing the same problem?
- How long has the problem been occurring?
- What changes have been made recently to the system? Were these changes initiated by you or via another support request?

Perform backups

- Backup all critical and important information
- Verify customer or client has done so if they are responsible for the system

Establish and Test a Theory

- Conduct research
  - Physically inspect the system
  - Reproduce the problem
  - Check system documentation or owner's manual
  - Ask other technicians for their thoughts; talk to previous technician who have worked on the system
  - Read vendor documentation and use the internet to conduct research for possible causes and solutions

Question the Obvious

1. Never overlook a simple solution such as ensuring the item is turned on or plugged in

2. Decide if the issue is likely hardware or software related

3. Determine if the issue physical or a logical

4. Test each theory individually to isolate the problem

Lesson Review: Establish a New Theory or Escalate

If initial theory does not check out, explore other causes and theories

Escalate the issue if it is:

- Beyond your skill level
- Beyond your knowledge level
- Out of scope per company policy or warranty documentation

Implement a Plan of Action

Repair, replace, workarounds

- Determine if solution to repair the system is cost effective
- In some cases, may be cheaper to replace
- Not all issues required to be resolved, in some cases a work around may be feasible

Implement the solution

- Apply the fix or replace the item
- If a workaround is to be put in place, ensure it will work correctly

Verify and Document

Verify solution resolves the issue

- Ensure that the system is fully functional
- Implement preventative measures to prevent reoccurrence

Document findings, actions, and outcomes.

- Parts used- cost and point of purchase information
- Steps taken- be as through as possible; more detail the better
- Tools used- software related or physical tools that were helpful to resolve the issue
- Lessons learned

Lesson Review: The Troubleshooting Methodology